Eric Cheung

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TECHNICAL SKILLS

Programming Languages:	Python, R, Go, Haskell, Prolog, Racket, C, C++, JavaScript, Swift, Kotlin, Matlab
Data Science & Machine Learning:	Pandas, NumPy, Scikit-Learn (Pipelines, PCA, k-Means, Logistic Regression, Naive Bayes, k-NN, Decision Trees, Random Forests, MLPs, Cross-Validation), PySpark, SciPy (stats, signal), Statsmodels (OLS, ANOVA, Tukey's HSD), Hypothesis Testing (t-tests, Mann-Whitney U, Chi-square), Kalman Filtering (pykalman), Matplotlib
Frontend Development:	Git, Next.js, React.js, React Native, Tailwind CSS
Backend Development:	Node.js, MongoDB, PostgreSQL, REST APIs, Postman, OAuth, GraphQL, Firebase
DevOps:	Bash, Docker, Ansible, Puppet, GitHub Actions, AWS EC2
System Administration:	Linux (Ubuntu, Kali, CentOS), Active Directory, Domain Controller, Group Policy, DNS, DHCP, TCP/IP, Wireshark, Jira, Confluence, VMware, VirtualBox
Software Engineering:	SDLC, Requirements Analysis, Architectural Design, UML, Agile Methodologies (Scrum, Kanban)

TECHNICAL WORK EXPERIENCE

Systems Support Technician June 2022 - June 2024

Framestore Studio, Vancouver, BC

- Operated with high autonomy under a remote manager (Montreal office), delivering rapid on-site and remote technical support across 2,000+ Jira tickets for floor-wide workstations and the studio's private screening theater.
- Provisioned and deployed 200+ production workstations from the ground up, managing thin-client and server imaging, physical setup, and seamless integration into the studio network.
- Standardized laptop imaging and deployment pipelines, ensuring rapid, secure hardware provisioning for incoming staff across macOS, Windows, and Linux.
- Orchestrated IT onboarding and technical enablement, guiding new hires through secure VPN authentication, access protocols, and standard workstation configurations.
- Automated routine administrative and troubleshooting workflows via custom Bash scripts, cutting ticket resolution time for application errors and optimizing access provisioning for production employees.
- Awarded the Framestore Staff Award (Spring 2023) for outstanding user support, cross-functional communication, and operational efficiency.

Freelance Web Developer October 2022 – March 2023

Miki S. Tattoo Art, Vancouver, BC

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- Developed an SEO-optimized web application using Next.js and Tailwind CSS, achieving a first-page Google ranking for “Vancouver Fineline Tattoo” within 3 months and increasing client inquiries by 30%.
- Designed a responsive UI with HTML, CSS, and JavaScript, incorporating client feedback to enhance engagement.
- Maintained site performance by implementing regular updates and bug fixes.

Laboratory Biologist January 2014 - August 2018

Nautilus Environmental, Burnaby, BC

- Conducted standardized environmental testing on species including: *Oncorhynchus mykiss, Pimephales promelas, Daphnia magna*, *Ceriodaphnia dubia*, *Hyalella azteca*, *Chironomus dilutus* and *Pseudokirchneriella subcapitata*, performing statistical analyses and reporting results for regulatory compliance and risk assessments.
- Trained new laboratory staff on standardized toxicity testing protocols.
- Contributed as a participating laboratory technician to multiple Environment and Climate Change Canada (ECCC) inter-laboratory validation studies; name acknowledged in the following official reference methods:
    - Biological Test Method: Toxicity Tests Using Aquatic Life Stages of Frogs (*Lithobates pipiens*) – STB 1/RM/62 (2024)
    - Biological Test Method: Reference Method for Determining Acute Lethality Using *Acartia tonsa* – STB 1/RM/60 (June 2019)
    - Biological Test Method: Reference Method for Determining Acute Lethality Using Threespine Stickleback (*Gasterosteus aculeatus*) – EPS 1/RM/10, Second Edition (December 2017)

TECHNICAL PROJECT EXPERIENCE

Unsupervised Pangenomic Analysis Pipeline in Python May – August 2026

CMPT 353 – Computational Data Science

- Architected an unsupervised machine learning pipeline in Python to analyze 62 *P. polymyxa* draft genomes, resolving ambiguous taxonomic boundaries and identifying evolutionary sub-lineages.
- Engineered a custom data parsing module with Pandas to extract Coding Sequences (CDS) from raw GFF3 files, filtering out biological noise to compile a 62 X 4,460 binary presence/absence feature matrix.
- Implemented Principal Component Analysis (PCA), K-Means, and Hierarchical Agglomerative Clustering (HCA) using Scikit-Learn to map structural genomic variance and visualize lineage dendrograms.
- Extracted specific genetic drivers of strain divergence by calculating absolute principal component loading weights, isolating the top gene biomarkers.

Reliable Data Transfer Protocol over UDP in Python September - December 2025

CMPT 371 - Data Communications and Networking

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- Implemented a pipelined RDT protocol over UDP with cumulative ACKs and fast retransmission, achieving 100% in-order delivery across all proxy-simulated test scenarios including loss, delay, duplication, and bit-error injection.
- Designed TCP-style congestion control and receiver-advertised flow control, maintaining stable throughput under low-consume conditions by dynamically adapting cwnd and rwnd behavior.
- Validated protocol through a comprehensive test suite, confirming proper behavior of handshake, retransmission, buffer management, and congestion recovery across all required invariants.

Reservation Management System in C++ May - August 2025

CMPT 276 - Introduction to Software Engineering

- Led a three-person team through the SDLC, architecting a C++ console-based reservation system that met project requirements.
- Authored comprehensive project documentation, including requirements and user manuals, ensuring system maintainability and supporting future scalability.
- Tested the system using unit and black-box testing methods, resolving 95% of bugs and enhancing system reliability.

EDUCATION

Bachelor of Science in Computer Science September 2024 – Present

Simon Fraser University
8888 University Drive, Burnaby, BC V5A 1S6
CGPA: 3.99 / 4.33

Computer Systems Technology Diploma April 2020 – April 2022

Vancouver Community College
1155 East Broadway, Vancouver, BC V5T 4V5
CGPA: 3.98 / 4.33

Bachelor of Science in Molecular Biology and Biochemistry September 2009 – April 2014

Simon Fraser University
8888 University Drive, Burnaby, BC V5A 1S6

AWARDS & CERTIFICATIONS

Framestore Studio – Staff Award Spring 2023
Cisco Networking Academy - NDG Linux Essentials Certificate Completed December 2021